Designing a 3D Printed Boat from a Prompt

A plain language report on asking for an innovative boat and then checking whether it actually floats and holds together

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What this report covers

This is the second part of the same small project. I built a helper that lets me describe a part in plain words and have it modelled in FreeCAD. The helper is an MCP, a bridge that lets the AI talk to the CAD program and run commands. I will not spend time on how the bridge works. I wanted to see how far it could go, so instead of a normal part I asked it for something hard and open ended: a unique, innovative boat. Then I checked the result myself with hand calculation and with simulation in ANSYS.

The boat is small, about 2.2 metres long. It is meant to be made by large scale 3D printing, printed standing on its back end, and it has to carry two people and a small outboard motor on a lake. The interesting part is that the printing method is awkward, and a good design has to work around that. I wanted to see if the helper could invent a design that respects those limits, not just draw a boat shape.

The result in one line: the helper invented a genuinely clever design, worked out its own weight and balance, and the three things I could check independently, the way it floats, its stability, and the strength of the motor mount, all passed.

Step 1. The request I gave

I started with almost nothing, on purpose, to see what it would do with a vague brief:

> Now I want you to build a boat. The concept is that it has to be unique and innovative.

It refused to guess the things that really matter and asked me for them instead. So I gave it the real constraints:

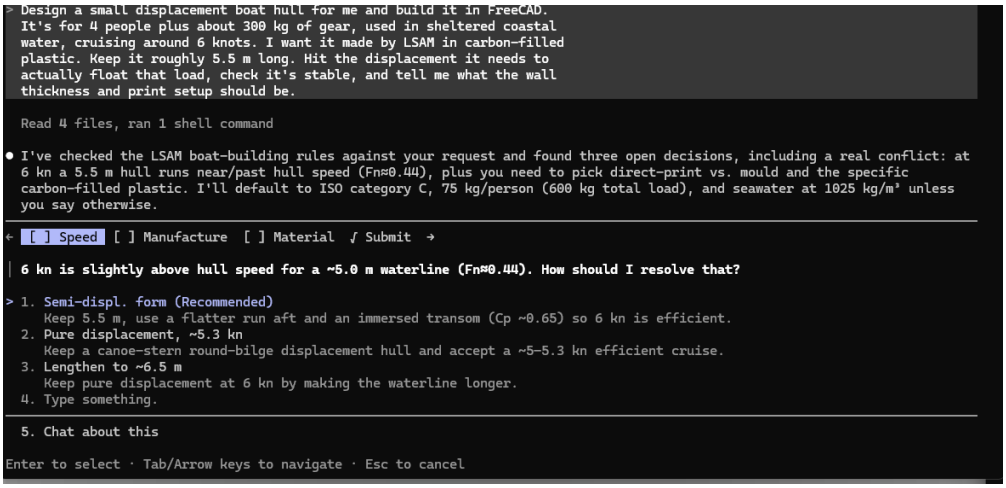
> The boat has a maximum length of 2 to 2.2 m. The process is large scale 3D printing and it is printed upright, meaning lengthwise. It must seat two people, about 150 kg, plus a motor around 30 kg at the back. The back needs more thickness to hold the motor, about 20 cm of it.
>
> The back where the motor sits only needs the 20 cm, the rest you can decide. The nozzle is 6 mm and the preferred wall is 12 mm, so two beads, and the back can be four beads. The material is PETG. The print time is roughly 20 hours. No overhangs steeper than 45 degrees. It is for a lake. No need for high speed, 4 to 6 knots is fine. Make something unique as you wish and show me the result.

Step 2. How the helper thought about it

This is where it impressed me. Printing a boat standing on its end creates two real problems, and the helper spotted both. First, the layers are stacked across the length of the boat, so any bending along the length tries to pull the layers apart, which is the weak direction for a printed part. Second, anything flat and crosswise becomes an overhang that cannot be printed without support.

Rather than hide these problems, it turned them into the design. It called the boat the Chevron, and its main ideas were:

- Sloped seat boxes. Every crosswise surface is set at 45 degrees so it prints cleanly with no support. The seat boxes double as sealed air tanks that keep the boat afloat even if it is holed.
- Tensioned tendons. Full length channels run along the keel, the chines and the gunwales. You thread steel wire rope through them and tension it, which squeezes the printed layers together so they cannot pull apart under bending. This fixes the weak direction head on.
- A flat backed hull shape suited to the slow lake speeds I asked for.

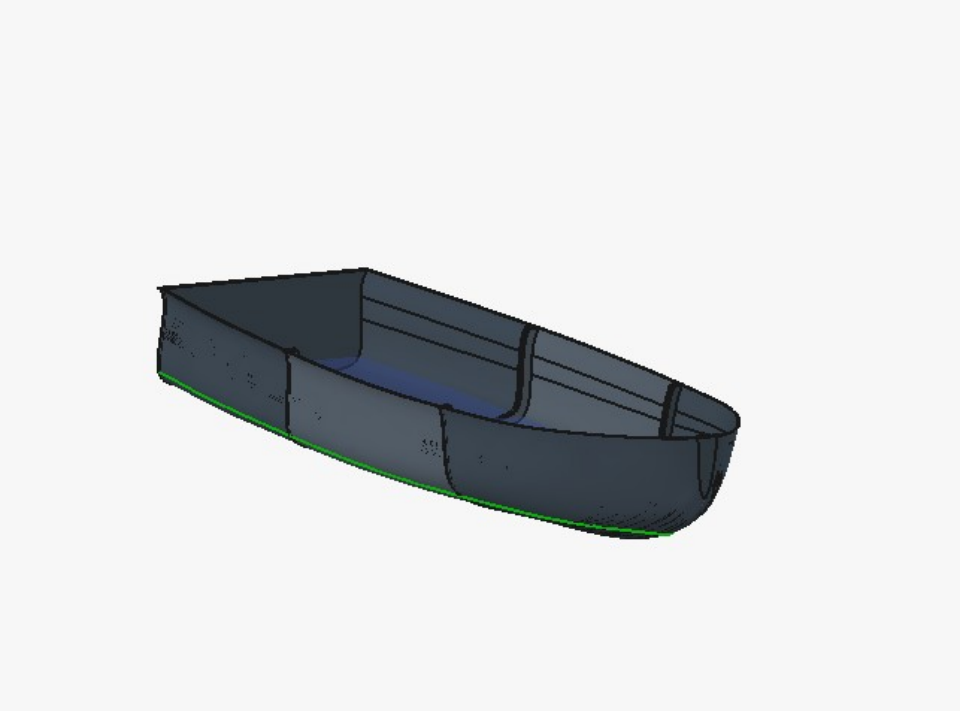


The helper gathering the few things it would not guess before starting, such as where the boat is used and how it is loaded.

It then built the hull, measured it, found it too heavy, and reworked the walls and ribs several times on its own until the weight and the overhang rules were both satisfied. It was open about getting its first weight estimate wrong and explaining why.

Step 3. The boat it built

The final design is one solid piece that prints as a single job with nothing to cut or drill afterwards, and no surface steeper than 45 degrees. A lighter version came out at about 94 kg, carrying two people and the motor.



The Chevron hull in FreeCAD. The pointed bow is on the right, the flat motor transom on the left, with the sloped seat boxes and internal ribs visible.

Item	Value
Length, beam, depth	2200 x 1283 x 601 mm
Hull weight (lighter version)	about 94 kg of PETG
Loaded weight (hull, 2 people, motor)	about 274 kg
Draft and trim	211 mm, sitting level
Wall thickness	12 mm normal, 24 mm at the motor mount
Print	single job, standing on the transom, no supports

As with any complex curved surface made this way, the hull is a concept shape that would still need smoothing by eye before it was built for real. The helper said this clearly and did not oversell it.

Step 4. Checking the result for real

The helper reported a lot of confident numbers. A professional looking table is not proof, so I picked the three claims that can actually be checked and checked them myself. Two were quick hand and geometry checks. The third needed simulation.

Check 1. Does it float as claimed

This is simple physics. The weight of water the hull pushes aside has to equal the weight of the loaded boat. I had the helper measure the underwater volume of the hull at the waterline it claimed, directly from the model, and compared it to the loaded weight.

● The underwater volume at **211 mm draft is 0.27509 m³**. I measured it on the outer hull shape: everything below a level plane at z = 211 mm, which came out as one valid solid.

Submerged volume	**275,093,245 mm³ = 0.27509 m³**
Displacement, fresh water (1000 kg/m³)	**275.1 kg**
Displacement, salt water (1025 kg/m³)	282.0 kg
Centre of buoyancy	784.4 mm forward of the transom (LCB), 133.4 mm above the keel (KB)
Waterplane	1.971 m², 1939 × 1175 mm

That's within 1 kg (0.4%) of the 274.1 kg loaded weight, which confirms the draft I solved for earlier. The extra 1 kg is because the earlier solve rounded the draft to 211 mm and had 0.07° of trim, while this cut is perfectly level. The centre of buoyancy at 784 mm also lines up with the centre of gravity at 786 mm, so the boat floats level.

Measured underwater volume at the 211 mm waterline: 0.27509 cubic metres, which is 275 kg of fresh water.

The underwater volume gives 275 kg of buoyancy against a loaded weight of 274 kg. That is a match within less than half a percent. The boat floats as claimed, confirmed by independent measurement, not just stated.

Check 2. Is it stable

Stability depends on the shape of the boat at the waterline. I had the model work out this shape property two different ways and compared them. The two methods agreed to within 1.6 percent, which confirms the stability figure. The boat comes out very stable for its size, which makes sense for a wide, flat lake dinghy.

The two methods agree to within 1.6%. The FreeCAD value is the more accurate one; the slice method slightly over-counts near the rounded ends of the waterplane.

Comparison with 673 mm:
- **v1 Chevron:** loaded at 293.7 kg, draft 222 mm, displaced volume 0.2937 m³, so BM = **673 mm**.
- **Chevron_Lite:** loaded at 274.1 kg, draft 211 mm, displaced volume 0.27509 m³, so BM = **697 mm**. That's **24 mm (+3.6%) more** than v1.

The reason is that the Lite boat sits 11 mm higher in the water, so it displaces 6.3% less water. The waterplane barely changes over those 11 mm, because the hull sides above the chine are nearly straight, so I_T drops only slightly. BM is I_T divided by volume, so it goes up.

Combined with KB = 133 mm and KG = 384 mm, this gives **GM ≈ 446 mm**, the value I reported for Lite. That's a stiff, very stable boat for a 2.2 m lake dinghy.

Two independent ways of working out the stability shape factor agree closely, confirming the stability number.

The one honest caveat here is that the final stability figure also depends on where the weight sits inside the boat, and that part is an assumption, not a measurement. Even so, the margin is so large that the boat stays clearly stable.

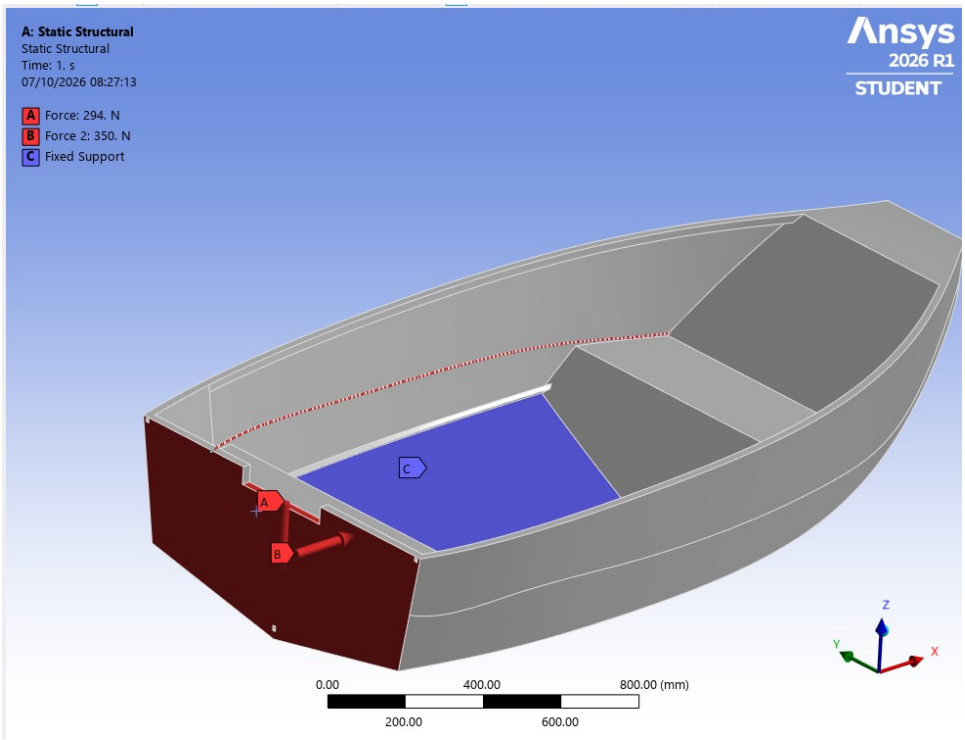
Check 3. Is the motor mount strong enough

This is the part most likely to fail, so it earned a full simulation. I exported the hull, set the plastic material to PETG, and loaded the back the way a motor really loads it: its weight pressing down, about 294 N, and its thrust pushing forward, about 350 N.

Properties of Outline Row 3: PETG

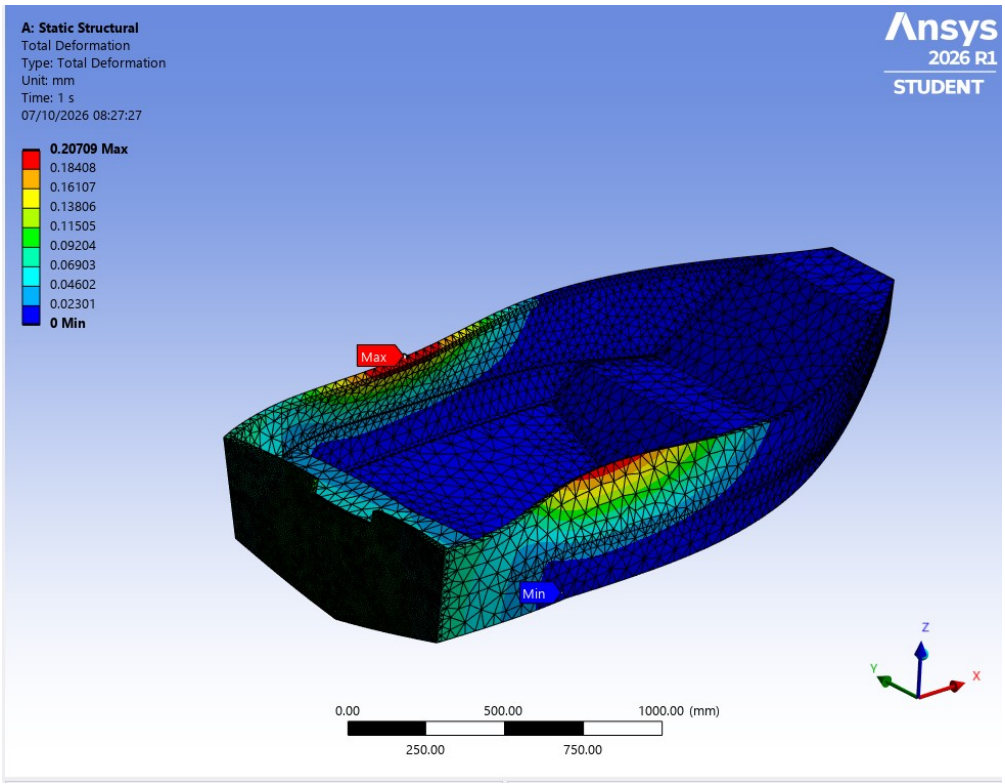
	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	1270	kg m^-3		
4	Isotropic Elasticity				
5	Derive from	Young's Modulus an...			
6	Young's Modulus	2000	MPa		
7	Poisson's Ratio	0.4			
8	Bulk Modulus	3.3333E+09	Pa		
9	Shear Modulus	7.1429E+08	Pa		
10	Tensile Yield Strength	50	MPa		

The PETG plastic set up in ANSYS. A soft material compared to steel, so the material choice matters here.

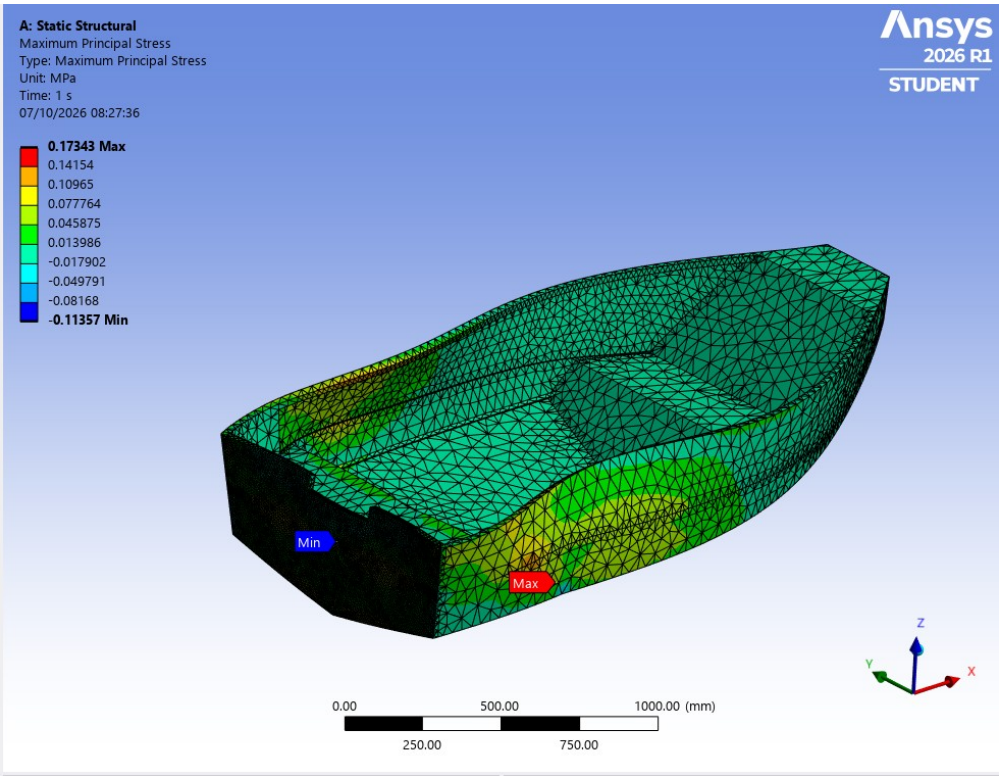


The two motor loads applied at the transom, with the cockpit floor held so the load has to travel through the hull.

Working out how to hold a floating boat in a simulation took a couple of attempts, because a boat is not bolted down anywhere. Once it was set up properly, holding the floor and loading the transom, the result was clear.



How much the loaded boat flexes. The most movement is about 0.2 mm at the top of the transom, which is very small.



Stress in the hull under the motor loads. The peak near the motor mount is only about 0.17 MPa.

The stress at the motor mount is tiny, far below what PETG can take, and the flexing is a fraction of a millimetre. The 24 mm reinforced back is more than strong enough for a 30 kg motor. The number is low partly because a thick plastic panel barely notices such a small load, which is a real result, not an error.

What was checked and what was not

As with the gear, here is the honest line.

Checked:

- Flotation. The measured underwater volume matches the loaded weight within half a percent.
- Stability shape. Confirmed two independent ways to within 1.6 percent, with the weight position noted as an assumption.
- Motor mount strength. Confirmed by simulation, with a large margin.

Not checked, and these matter:

- Whether the single bead walls of the air tanks will be watertight. This cannot be answered by simulation at all. It can only be found by printing a sample and testing it with water or air pressure.
- Whether the tensioned tendon idea really stops the layers pulling apart. It is a clever idea, but it is still an idea that needs a physical test.
- Bending of the whole hull in waves, and a formal boat stability and structure certification to the proper standards.

Conclusion

Given a one line request for an innovative boat, the helper produced a design that is genuinely thoughtful. It did not just draw a boat. It understood how the printing method would fight the design and built the whole concept around solving that. It then worked out its own weight, balance and stability, corrected itself along the way, and was honest about what still needed testing.

When I checked the parts that can be checked, the boat floated as claimed, was stable, and its motor mount was strong. The parts that remain open are the ones that honestly can only be settled by printing a real one and testing it, above all whether the thin tank walls stay dry.

So the right label for this boat is a checked concept, not a finished product. That is still a strong outcome. The helper took a vague idea to a real, measured, partly proven design in one sitting, which is exactly what I was hoping to find out it could do.